

SOC-01: SSH Invalid-User Login Detection

Brute-Force / Credential-Access Hunting with Wazuh SIEM

Lab ID	SOC-01
Category	SOC / Threat Detection & Hunting
Platform	Wazuh 4.x (Indexer + Dashboard)
Data Source	journald / sshd-session logs
Monitored Host	monitor (agent.id 001, 192.168.122.187)
Manager	mike-pc
Attacker Source	192.168.122.97
MITRE ATT&CK	T1110.001 (Password Guessing), T1021.004 (SSH)
Tactics	Credential Access, Lateral Movement
Analyst	Michael.A / SafeTest-Dev
Date	Jun 08, 2026

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1. Executive Summary

This lab documents the detection and triage of an SSH credential-access attempt against a monitored Linux host using the Wazuh SIEM. During a 15-minute window on June 8, 2026, the Wazuh agent on host **monitor** generated a burst of authentication-failure alerts. The events were dominated by repeated attempts to authenticate over SSH using the non-existent username **hacker** originating from the single source address **192.168.122.97**.

The activity maps to MITRE ATT&CK techniques **T1110.001 (Brute Force: Password Guessing)** and **T1021.004 (Remote Services: SSH)**, and escalated through Wazuh's correlation engine from individual level-5 alerts (rule 5710) to a level-10 composite alert (rule 2502) once the failure threshold was exceeded within the time window.

No successful authentication was observed. The recommended response is source-IP containment, review of SSH exposure, and enforcement of key-based authentication.

2. Lab Environment

SIEM	Wazuh 4.x — Indexer, Manager and Dashboard
Index	wazuh-alerts-4.x-2026.06.07
Agent name	monitor
Agent ID	001
Agent IP	192.168.122.187
Manager	mike-pc
Decoder	sshd
Log source	journald (input.type: log)
Program	sshd-session

Wazuh ingests systemd journal entries from the SSH daemon, decodes them with the built-in sshd decoder, and matches them against ruleset signatures. The Threat Hunting / Discover views in the Wazuh Dashboard were used to pivot, filter and inspect individual alert documents.

3. Detection & Event Timeline

The Threat Hunting dashboard was scoped to `manager.name: mike-pc` and `agent.id: 001`. The events histogram showed a steady stream of agent activity with a pronounced spike around 01:07, where the SSH authentication failures clustered. A total of 189 hits were recorded across the 15-minute baseline window.

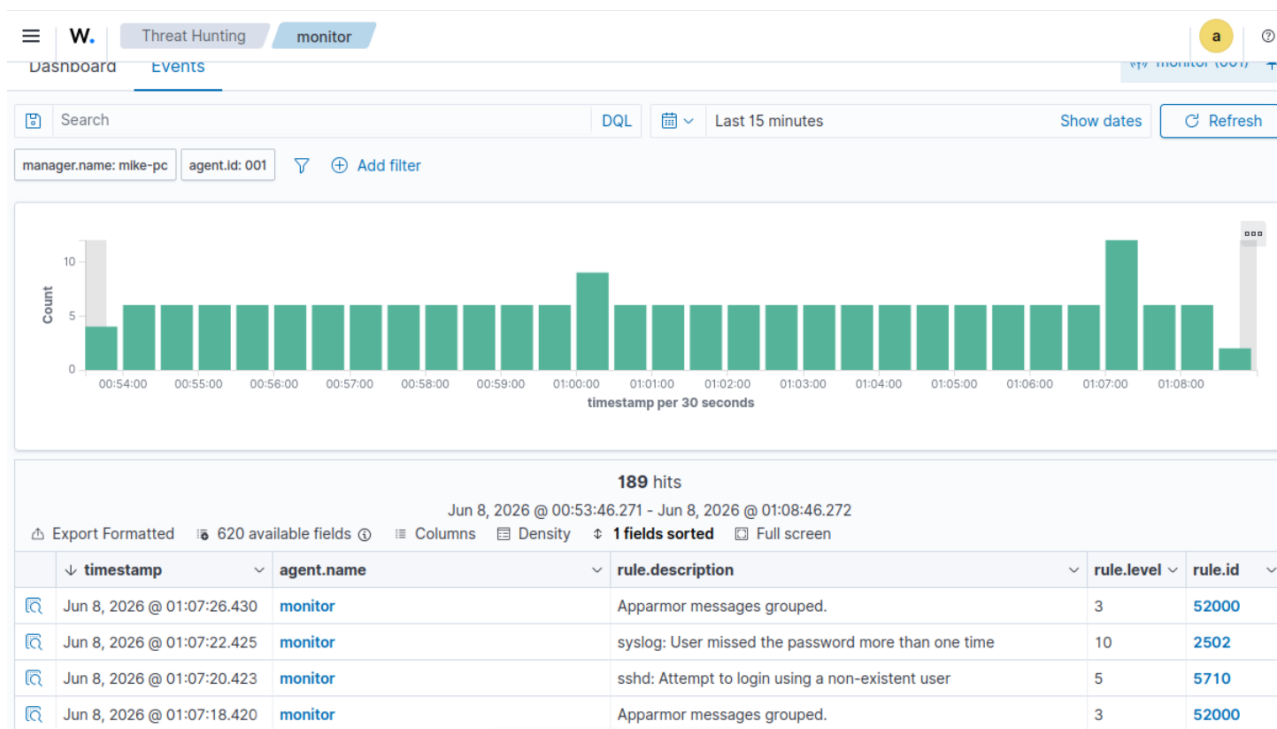


Figure 1 — Threat Hunting / Events view: alert histogram and the alert table showing rule 5710 (invalid user, level 5) and rule 2502 (repeated password miss, level 10).

Filtering to the relevant rule activity isolated **6 hits** within a narrow 01:01–01:16 window, all from agent **monitor**. The sequence shows repeated `sshd: Attempt to login using a non-existent user` (rule 5710, level 5) events, a `PAM: User login failed` event (rule 5503), and finally the correlated `syslog: User missed the password more than one time` alert (rule 2502, level 10).

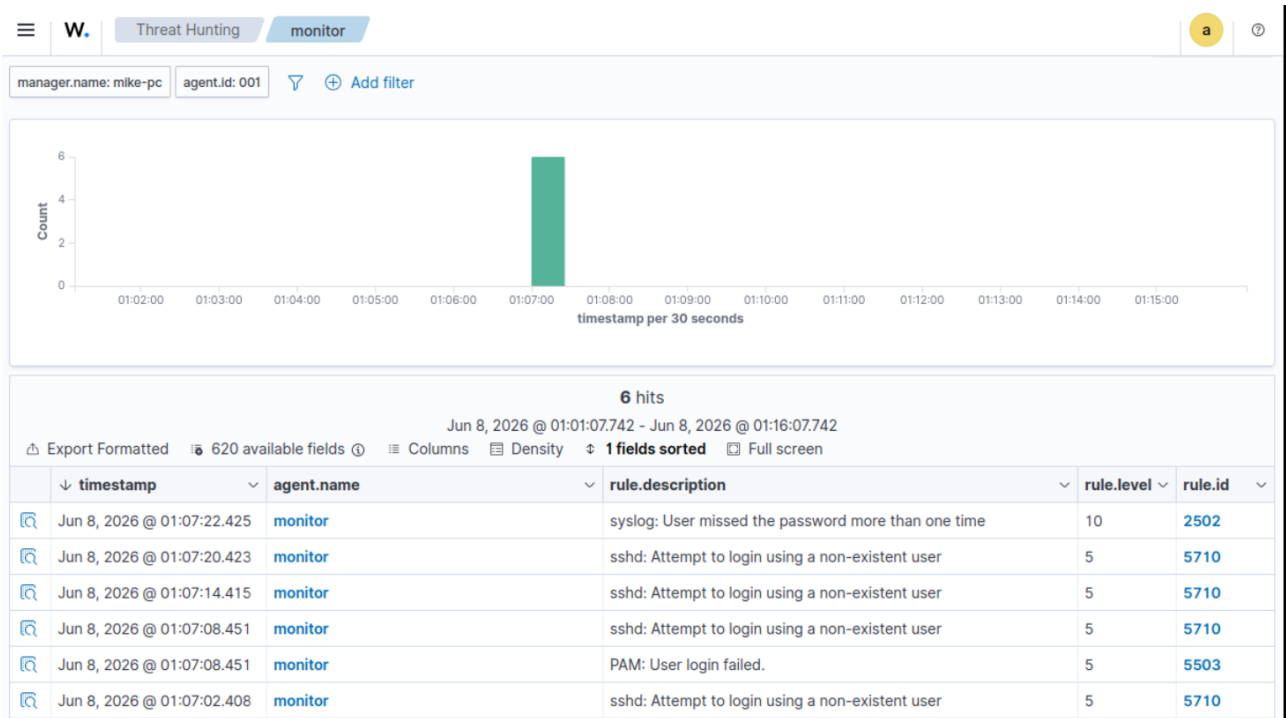


Figure 2 — Filtered detection: 6 correlated SSH authentication-failure events culminating in the level-10 escalation alert.

4. Primary Alert — Rule 5710 (Invalid User)

The seed event of this incident is Wazuh rule **5710**: an attempt to authenticate over SSH using a username that does not exist on the host. The raw log line captured by the sshd decoder is shown below.

```
Jun 07 18:07:20 monitor sshd-session[86982]: Failed password
for invalid user hacker from 192.168.122.97 port 58318 ssh2
```

Expanding the alert document in the Discover view exposes the decoded fields, including the attacker-controlled username and source IP, the MITRE mapping, and the compliance tags.

W.

Discover

wazuh-alerts-4.x-2026.06.07#eApEo54B-is50HBcm8xG

a

Table

JSON

@timestamp	Jun 8, 2026 @ 01:07:20.423
_index	wazuh-alerts-4.x-2026.06.07
agent.id	001
agent.ip	192.168.122.187
agent.name	monitor
data.srcip	192.168.122.97
data.srcuser	hacker
decoder.name	sshd
decoder.parent	sshd
full_log	Jun 07 18:07:20 monitor sshd-session[86982]: Failed password for invalid user hacker from 192.168.122.97 port 58318 ssh2
id	1780855640.181251
input.type	log
location	journalld
manager.name	mike-pc
predecoder.hostname	monitor
predecoder.program_name	sshd-session
predecoder.timestamp	Jun 07 18:07:20
rule.description	sshd: Attempt to login using a non-existent user

Figure 3 — Document detail for rule 5710. Note `data.srcuser = hacker` and `data.srcip = 192.168.122.97` extracted from the raw log.

Key fields — Rule 5710

rule.id	5710
rule.level	5
rule.description	sshd: Attempt to login using a non-existent user
data.srcuser	hacker
data.srcip	192.168.122.97
rule.groups	syslog, sshd, authentication_failed, invalid_login
MITRE technique	Password Guessing (T1110.001), SSH (T1021.004)
MITRE tactic	Credential Access, Lateral Movement
timestamp	Jun 8, 2026 @ 01:07:20.423

5. Escalation Alert — Rule 2502 (Repeated Failures)

As the invalid-user attempts repeated within the correlation window, Wazuh fired rule **2502** — *syslog: User missed the password more than one time* — at **level 10**. This composite alert is the clearest signal of a deliberate brute-force / password-guessing effort rather than an isolated typo, and is the alert an analyst should prioritise.

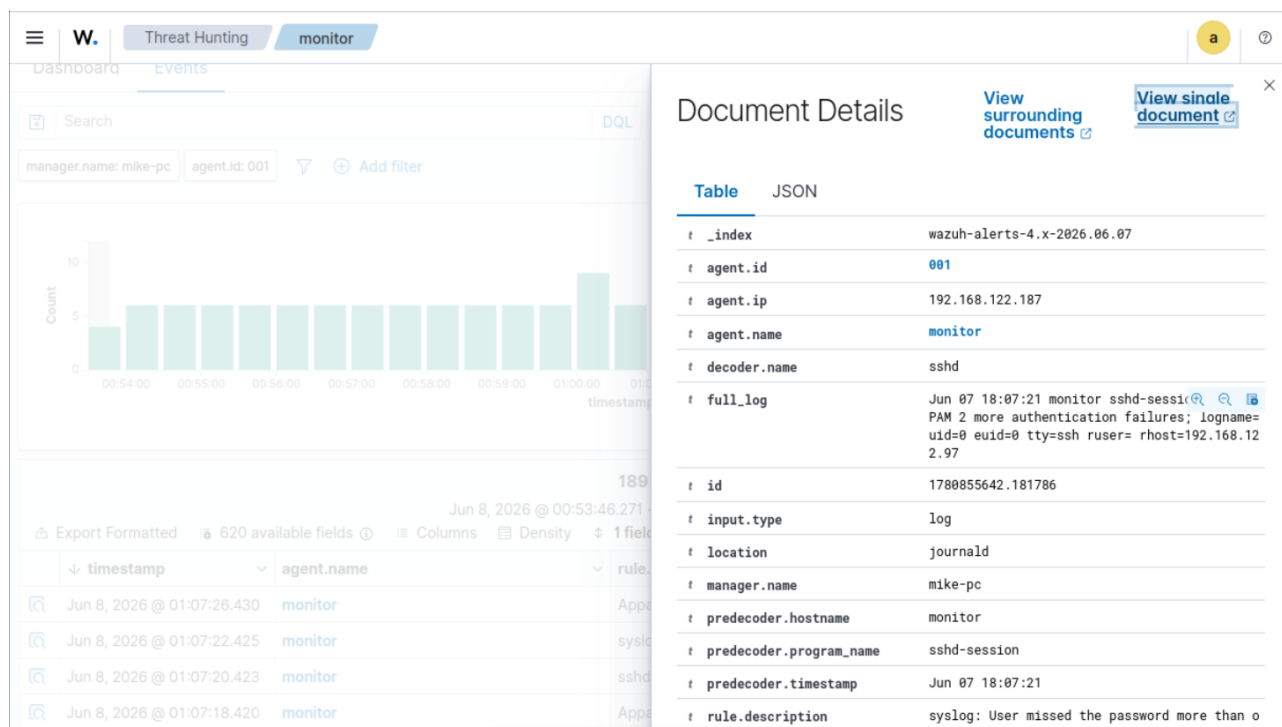


Figure 4 — Document Details panel for the level-10 escalation alert (rule 2502), showing the PAM authentication-failure log and decoded metadata.

The full table view of the same document enumerates every decoded field, including the PAM log line that records multiple authentication failures from the same remote host.

Table	JSON
@timestamp	Jun 8, 2026 @ 01:07:22.425
_index	wazuh-alerts-4.x-2026.06.07
agent.id	001
agent.ip	192.168.122.187
agent.name	monitor
decoder.name	sshd
full_log	Jun 07 18:07:21 monitor sshd-session[86982]: PAM 2 more authentication failures; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.122.97
id	1780855642.181786
input.type	log
location	journald
manager.name	mike-pc
predecoder.hostname	monitor
predecoder.program_name	sshd-session
predecoder.timestamp	Jun 07 18:07:21
rule.description	syslog: User missed the password more than one time
rule.firedtimes	1
rule.gdpr	IV.35.7.d, IV.32.2

Figure 5 — Decoded field table for rule 2502 (top half): index, agent, decoder, full_log and rule description.

6. Rule Metadata, MITRE & Compliance Mapping

The lower half of the rule 2502 document lists the regulatory and framework mappings Wazuh attaches to the alert. These tags allow the detection to be tied directly to audit and compliance reporting obligations.

W.	Discover	wazuh-alerts-4.x-2026.06.07#eQpEo54B-is50HBcpsz_	a	?
t	predecoder.hostname	monitor		
t	predecoder.program_name	sshd-session		
t	predecoder.timestamp	Jun 07 18:07:21		
t	rule.description	syslog: User missed the password more than one time		
#	rule.firedtimes	1		
t	rule.gdpr	IV_35.7.d, IV_32.2		
t	rule.gpg13	7.8		
t	rule.groups	syslog, access_control, authentication_failed		
t	rule.hipaa	164.312.b		
t	rule.id	2502		
#	rule.level	10		
🔍	rule.mail	false		
t	rule.mitre.id	T1110		
t	rule.mitre.tactic	Credential Access		
t	rule.mitre.technique	Brute Force		
t	rule.nist_800_53	AU.14, AC.7		
t	rule.pci_dss	10.2.4, 10.2.5		
t	rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3		
📅	timestamp	Jun 8, 2026 @ 01:07:22.425		

Figure 6 — Compliance and MITRE metadata for rule 2502 (level 10): GDPR, HIPAA, PCI-DSS, NIST 800-53, TSC and MITRE Brute Force (T1110).

Compliance summary

rule.id / level	2502 / 10 (escalation)
MITRE (5710)	T1110.001, T1021.004
MITRE (2502)	T1110 – Brute Force / Credential Access
PCI-DSS	10.2.4, 10.2.5, 10.6.1
HIPAA	164.312.b
NIST 800-53	AU.14, AC.7, AU.6
GDPR	IV_35.7.d, IV_32.2
TSC	CC6.1, CC6.8, CC7.2, CC7.3
GPG13	7.1 / 7.8

7. Analysis & Findings

Attack characterisation

A single external source (192.168.122.97) issued repeated SSH authentication requests against host monitor using the username **hacker**, which does not exist on the system. The use of a generic, non-existent account name and the rapid repetition within seconds are consistent with automated password guessing rather than legitimate user error.

Detection chain

1) Each invalid-user attempt independently triggered rule 5710 (level 5). 2) A PAM failure surfaced as rule 5503. 3) Once the failures crossed Wazuh's frequency threshold within the correlation window, the composite rule 2502 fired at level 10, raising the incident to an alertable severity. This demonstrates Wazuh's value: low-severity primitives are aggregated into a high-confidence, high-severity signal.

Impact assessment

No Accepted password or successful-session event accompanied the failures, indicating the attempts did **not** succeed. Risk remains elevated while the SSH service is reachable from the source network and the attacker can continue guessing.

8. Recommendations & Response

Containment	Block / rate-limit 192.168.122.97 at the host firewall and perimeter; enable fail2ban or Wazuh active-response to auto-ban repeat offenders on rule 2502.
Hardening	Disable password authentication and enforce SSH key-based auth (PasswordAuthentication no); disable direct root login (PermitRootLogin no).
Exposure	Restrict SSH to a management VLAN / VPN; do not expose port 22 to untrusted networks; consider port-knocking or a non-default port to reduce automated noise.
Detection	Tune/confirm Wazuh correlation thresholds for rule 2502; alert SOC on any level ≥ 10 SSH alert; add a watchlist for the source IP.
Hunting	Pivot on data.scrip across all agents to confirm the source is not targeting other hosts; review for any successful logins from the same subnet.